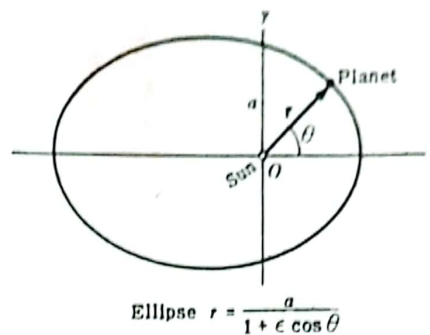


From analytic geometry, the polar equation of a conic section with focus at the origin and eccentricity ϵ is $r = \frac{a}{1 + \epsilon \cos \theta}$ where a is a constant. Comparing this with the equation derived, it is seen that the required orbit is a conic section with eccentricity $\epsilon = p/GM$. The orbit is an ellipse, parabola or hyperbola according as ϵ is less than, equal to or greater than one. Since orbits of planets are closed curves it follows that they must be ellipses.



LINE INTEGRALS

6. If $A = (3x^2 + 6y)\mathbf{i} - 14yz\mathbf{j} + 20xz^2\mathbf{k}$, evaluate $\int_C A \cdot d\mathbf{r}$ from $(0,0,0)$ to $(1,1,1)$ along the following paths C :

- $x = t, y = t^2, z = t^3$.
- the straight lines from $(0,0,0)$ to $(1,0,0)$, then to $(1,1,0)$, and then to $(1,1,1)$.
- the straight line joining $(0,0,0)$ and $(1,1,1)$.

$$\begin{aligned} \int_C A \cdot d\mathbf{r} &= \int_C [(3x^2 + 6y)\mathbf{i} - 14yz\mathbf{j} + 20xz^2\mathbf{k}] \cdot (dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}) \\ &= \int_C (3x^2 + 6y) dx - 14yz dy + 20xz^2 dz \end{aligned}$$

(a) If $x = t, y = t^2, z = t^3$, points $(0,0,0)$ and $(1,1,1)$ correspond to $t = 0$ and $t = 1$ respectively. Then

$$\begin{aligned} \int_C A \cdot d\mathbf{r} &= \int_{t=0}^1 (3t^2 + 6t^2) dt - 14(t^2)(t^3) d(t^2) + 20(t)(t^3)^2 d(t^3) \\ &= \int_{t=0}^1 9t^2 dt - 28t^8 dt + 60t^9 dt \\ &= \int_{t=0}^1 (9t^2 - 28t^8 + 60t^9) dt = 3t^3 - 4t^7 + 6t^{10} \Big|_0^1 = 5 \end{aligned}$$

Another Method.

Along C , $A = 9t^2\mathbf{i} - 14t^5\mathbf{j} + 20t^7\mathbf{k}$ and $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}$ and $d\mathbf{r} = (1 + 2t\mathbf{j} + 3t^2\mathbf{k})dt$.

$$\begin{aligned} \text{Then } \int_C A \cdot d\mathbf{r} &= \int_{t=0}^1 (9t^2\mathbf{i} - 14t^5\mathbf{j} + 20t^7\mathbf{k}) \cdot (1 + 2t\mathbf{j} + 3t^2\mathbf{k}) dt \\ &= \int_0^1 (9t^2 - 28t^8 + 60t^9) dt = 5 \end{aligned}$$

(b) Along the straight line from $(0,0,0)$ to $(1,0,0)$ $y = 0, z = 0, dy = 0, dz = 0$ while x varies from 0 to 1. Then the integral over this part of the path is

$$\int_{x=0}^1 (3x^2 + 6(0)) dx - 14(0)(0)(0) + 20x(0)^2(0) = \int_{x=0}^1 3x^2 dx = x^3 \Big|_0^1 = 1$$

Along the straight line from $(1,0,0)$ to $(1,1,0)$ $x = 1, z = 0, dx = 0, dz = 0$ while y varies from 0 to 1. Then the integral over this part of the path is

$$\int_{y=0}^1 (3(1)^2 + 6y) 0 - 14y(0) dy + 20(1)(0)^2 0 = 0$$

Along the straight line from $(1,1,0)$ to $(1,1,1)$ $x=1, y=1, dx=0, dy=0$ while z varies from 0 to 1. Then the integral over this part of the path is

$$\int_{z=0}^1 (3(1)^2 + 6(1)) 0 - 14(1) z(0) + 20(1) z^2 dz = \int_{z=0}^1 20z^2 dz = \frac{20z^3}{3} \Big|_0^1 = \frac{20}{3}$$

Adding,
$$\int_C \mathbf{A} \cdot d\mathbf{r} = 1 + 0 + \frac{20}{3} = \frac{23}{3}$$

(c) The straight line joining $(0,0,0)$ and $(1,1,1)$ is given in parametric form by $x=t, y=t, z=t$. Then

$$\begin{aligned} \int_C \mathbf{A} \cdot d\mathbf{r} &= \int_{t=0}^1 (3t^2 + 6t) dt - 14(t)(t) dt + 20(t)(t)^2 dt \\ &= \int_{t=0}^1 (3t^2 + 6t - 14t^2 + 20t^3) dt = \int_{t=0}^1 (6t - 11t^2 + 20t^3) dt = \frac{13}{3} \end{aligned}$$

7. Find the total work done in moving a particle in a force field given by $\mathbf{F} = 3xy\mathbf{i} - 5z\mathbf{j} + 10x\mathbf{k}$ along the curve $x=t^2+1, y=2t^2, z=t^3$ from $t=1$ to $t=2$.

$$\begin{aligned} \text{Total work} &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C (3xy\mathbf{i} - 5z\mathbf{j} + 10x\mathbf{k}) \cdot (dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}) \\ &= \int_C 3xy dx - 5z dy + 10x dz \\ &= \int_{t=1}^2 3(t^2+1)(2t^2) d(t^2+1) - 5(t^3) d(2t^2) + 10(t^2+1) d(t^3) \\ &= \int_1^2 (12t^5 + 10t^4 + 12t^3 + 30t^2) dt = 303 \end{aligned}$$

8. If $\mathbf{F} = 3xy\mathbf{i} - y^2\mathbf{j}$, evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where C is the curve in the xy plane, $y=2x^2$, from $(0,0)$ to $(1,2)$.

Since the integration is performed in the xy plane ($z=0$), we can take $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$. Then

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C (3xy\mathbf{i} - y^2\mathbf{j}) \cdot (dx\mathbf{i} + dy\mathbf{j}) \\ &= \int_C 3xy dx - y^2 dy \end{aligned}$$

First Method. Let $x=t$ in $y=2x^2$. Then the parametric equations of C are $x=t, y=2t^2$. Points $(0,0)$ and $(1,2)$ correspond to $t=0$ and $t=1$ respectively. Then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{t=0}^1 3(t)(2t^2) dt - (2t^2)^2 d(2t^2) = \int_{t=0}^1 (6t^3 - 16t^5) dt = -\frac{7}{6}$$

Second Method. Substitute $y=2x^2$ directly, where x goes from 0 to 1. Then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{x=0}^1 3x(2x^2) dx - (2x^2)^2 d(2x^2) = \int_{x=0}^1 (6x^3 - 16x^5) dx = -\frac{7}{6}$$

Note that if the curve were traversed in the opposite sense, i.e. from $(1,2)$ to $(0,0)$, the value of the integral would have been $7/6$ instead of $-7/6$.

9. Find the work done in moving a particle once around a circle C in the xy plane, if the circle has center at the origin and radius 3 and if the force field is given by

$$\mathbf{F} = (2x - y + z)\mathbf{i} + (x + y - z^2)\mathbf{j} + (3x - 2y + 4z)\mathbf{k}$$

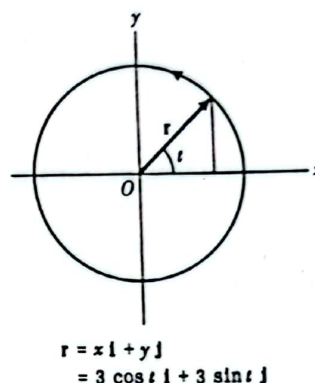
In the plane $z=0$, $\mathbf{F} = (2x - y)\mathbf{i} + (x + y)\mathbf{j} + (3x - 2y)\mathbf{k}$ and $d\mathbf{r} = dx\mathbf{i} + dy\mathbf{j}$ so that the work done is

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C [(2x - y)\mathbf{i} + (x + y)\mathbf{j} + (3x - 2y)\mathbf{k}] \cdot [dx\mathbf{i} + dy\mathbf{j}] \\ &= \int_C (2x - y)dx + (x + y)dy \end{aligned}$$

Choose the parametric equations of the circle as $x = 3 \cos t$, $y = 3 \sin t$ where t varies from 0 to 2π (see adjoining figure). Then the line integral equals

$$\begin{aligned} \int_{t=0}^{2\pi} [2(3 \cos t) - 3 \sin t] [-3 \sin t] dt + [3 \cos t + 3 \sin t] [3 \cos t] dt \\ = \int_0^{2\pi} (9 - 9 \sin t \cos t) dt = 9t - \frac{9}{2} \sin^2 t \Big|_0^{2\pi} = 18\pi \end{aligned}$$

In traversing C we have chosen the counterclockwise direction indicated in the adjoining figure. We call this the *positive* direction, or say that C has been traversed in the positive sense. If C were traversed in the clockwise (negative) direction the value of the integral would be -18π .



10. (a) If $\mathbf{F} = \nabla\phi$, where ϕ is single-valued and has continuous partial derivatives, show that the work done in moving a particle from one point $P_1 \equiv (x_1, y_1, z_1)$ in this field to another point $P_2 \equiv (x_2, y_2, z_2)$ is independent of the path joining the two points.

- (b) Conversely, if $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of the path C joining any two points, show that there exists a function ϕ such that $\mathbf{F} = \nabla\phi$.

$$\begin{aligned} \text{(a) Work done} &= \int_{P_1}^{P_2} \mathbf{F} \cdot d\mathbf{r} = \int_{P_1}^{P_2} \nabla\phi \cdot d\mathbf{r} \\ &= \int_{P_1}^{P_2} \left(\frac{\partial\phi}{\partial x} \mathbf{i} + \frac{\partial\phi}{\partial y} \mathbf{j} + \frac{\partial\phi}{\partial z} \mathbf{k} \right) \cdot (dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}) \\ &= \int_{P_1}^{P_2} \frac{\partial\phi}{\partial x} dx + \frac{\partial\phi}{\partial y} dy + \frac{\partial\phi}{\partial z} dz \\ &= \int_{P_1}^{P_2} d\phi = \phi(P_2) - \phi(P_1) = \phi(x_2, y_2, z_2) - \phi(x_1, y_1, z_1) \end{aligned}$$

Then the integral depends only on points P_1 and P_2 and not on the path joining them. This is true of course only if $\phi(x, y, z)$ is single-valued at all points P_1 and P_2 .

- (b) Let $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$. By hypothesis, $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of the path C joining any two points, which we take as (x_1, y_1, z_1) and (x, y, z) respectively. Then

$$\phi(x, y, z) = \int_{(x_1, y_1, z_1)}^{(x, y, z)} \mathbf{F} \cdot d\mathbf{r} = \int_{(x_1, y_1, z_1)}^{(x, y, z)} F_1 dx + F_2 dy + F_3 dz$$

is independent of the path joining (x_1, y_1, z_1) and (x, y, z) . Thus

$$\begin{aligned}
 \phi(x+\Delta x, y, z) - \phi(x, y, z) &= \int_{(x_1, y_1, z_1)}^{(x+\Delta x, y, z)} \mathbf{F} \cdot d\mathbf{r} - \int_{(x_1, y_1, z_1)}^{(x, y, z)} \mathbf{F} \cdot d\mathbf{r} \\
 &= \int_{(x, y, z)}^{(x+\Delta x, y, z)} \mathbf{F} \cdot d\mathbf{r} + \int_{(x_1, y_1, z_1)}^{(x, y, z)} \mathbf{F} \cdot d\mathbf{r} \\
 &= \int_{(x, y, z)}^{(x+\Delta x, y, z)} \mathbf{F} \cdot d\mathbf{r} = \int_{(x, y, z)}^{(x+\Delta x, y, z)} F_1 dx + F_2 dy + F_3 dz
 \end{aligned}$$

Since the last integral must be independent of the path joining (x, y, z) and $(x+\Delta x, y, z)$, we may choose the path to be a straight line joining these points so that dy and dz are zero. Then

$$\frac{\phi(x+\Delta x, y, z) - \phi(x, y, z)}{\Delta x} = \frac{1}{\Delta x} \int_{(x, y, z)}^{(x+\Delta x, y, z)} F_1 dx$$

Taking the limit of both sides as $\Delta x \rightarrow 0$, we have $\frac{\partial \phi}{\partial x} = F_1$.

Similarly, we can show that $\frac{\partial \phi}{\partial y} = F_2$ and $\frac{\partial \phi}{\partial z} = F_3$.

Then $\mathbf{F} = F_1 \mathbf{i} + F_2 \mathbf{j} + F_3 \mathbf{k} = \frac{\partial \phi}{\partial x} \mathbf{i} + \frac{\partial \phi}{\partial y} \mathbf{j} + \frac{\partial \phi}{\partial z} \mathbf{k} = \nabla \phi$.

If $\int_{P_1}^{P_2} \mathbf{F} \cdot d\mathbf{r}$ is independent of the path C joining P_1 and P_2 , then $\mathbf{F}$ is called a *conservative field*. It follows that if $\mathbf{F} = \nabla \phi$ then $\mathbf{F}$ is conservative, and conversely.

Proof using vectors. If the line integral is independent of the path, then

$$\phi(x, y, z) = \int_{(x_1, y_1, z_1)}^{(x, y, z)} \mathbf{F} \cdot d\mathbf{r} = \int_{(x_1, y_1, z_1)}^{(x, y, z)} \mathbf{F} \cdot \frac{d\mathbf{r}}{ds} ds$$

By differentiation, $\frac{d\phi}{ds} = \mathbf{F} \cdot \frac{d\mathbf{r}}{ds}$. But $\frac{d\phi}{ds} = \nabla \phi \cdot \frac{d\mathbf{r}}{ds}$ so that $(\nabla \phi - \mathbf{F}) \cdot \frac{d\mathbf{r}}{ds} = 0$.

Since this must hold irrespective of $\frac{d\mathbf{r}}{ds}$, we have $\mathbf{F} = \nabla \phi$.

11. (a) If $\mathbf{F}$ is a conservative field, prove that $\text{curl } \mathbf{F} = \nabla \times \mathbf{F} = \mathbf{0}$ (i.e. $\mathbf{F}$ is irrotational).
 (b) Conversely, if $\nabla \times \mathbf{F} = \mathbf{0}$ (i.e. $\mathbf{F}$ is irrotational), prove that $\mathbf{F}$ is conservative.

(a) If $\mathbf{F}$ is a conservative field, then by Problem 10, $\mathbf{F} = \nabla \phi$.
 Thus $\text{curl } \mathbf{F} = \nabla \times \nabla \phi = \mathbf{0}$ (see Problem 27(a), Chapter 4).

(b) If $\nabla \times \mathbf{F} = \mathbf{0}$, then

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} = \mathbf{0} \quad \text{and thus}$$

$$\frac{\partial F_3}{\partial y} = \frac{\partial F_2}{\partial z}, \quad \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x}, \quad \frac{\partial F_2}{\partial x} = \frac{\partial F_1}{\partial y}$$

We must prove that $\mathbf{F} = \nabla \phi$ follows as a consequence of this.

The work done in moving a particle from (x_1, y_1, z_1) to (x, y, z) in the force field $\mathbf{F}$ is

$$\int_C F_1(x,y,z) dx + F_2(x,y,z) dy + F_3(x,y,z) dz$$

where C is a path joining (x_1, y_1, z_1) and (x, y, z) . Let us choose as a particular path the straight line segments from (x_1, y_1, z_1) to (x, y_1, z_1) to (x, y, z_1) to (x, y, z) and call $\phi(x, y, z)$ the work done along this particular path. Then

$$\phi(x, y, z) = \int_{x_1}^x F_1(x, y_1, z_1) dx + \int_{y_1}^y F_2(x, y, z_1) dy + \int_{z_1}^z F_3(x, y, z) dz$$

It follows that

$$\frac{\partial \phi}{\partial z} = F_3(x, y, z)$$

$$\begin{aligned} \frac{\partial \phi}{\partial y} &= F_2(x, y, z_1) + \int_{z_1}^z \frac{\partial F_2}{\partial y}(x, y, z) dz \\ &= F_2(x, y, z_1) + \int_{z_1}^z \frac{\partial F_2}{\partial z}(x, y, z) dz \\ &= F_2(x, y, z_1) + F_2(x, y, z) \Big|_{z_1}^z = F_2(x, y, z_1) + F_2(x, y, z) - F_2(x, y, z_1) = F_2(x, y, z) \end{aligned}$$

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= F_1(x, y_1, z_1) + \int_{y_1}^y \frac{\partial F_1}{\partial x}(x, y, z_1) dy + \int_{z_1}^z \frac{\partial F_1}{\partial x}(x, y, z) dz \\ &= F_1(x, y_1, z_1) + \int_{y_1}^y \frac{\partial F_1}{\partial y}(x, y, z_1) dy + \int_{z_1}^z \frac{\partial F_1}{\partial z}(x, y, z) dz \\ &= F_1(x, y_1, z_1) + F_1(x, y, z_1) \Big|_{y_1}^y + F_1(x, y, z) \Big|_{z_1}^z \\ &= F_1(x, y_1, z_1) + F_1(x, y, z_1) - F_1(x, y_1, z_1) + F_1(x, y, z) - F_1(x, y, z_1) = F_1(x, y, z) \end{aligned}$$

Then $\mathbf{F} = F_1 \mathbf{i} + F_2 \mathbf{j} + F_3 \mathbf{k} = \frac{\partial \phi}{\partial x} \mathbf{i} + \frac{\partial \phi}{\partial y} \mathbf{j} + \frac{\partial \phi}{\partial z} \mathbf{k} = \nabla \phi.$

Thus a necessary and sufficient condition that a field $\mathbf{F}$ be conservative is that $\text{curl } \mathbf{F} = \nabla \times \mathbf{F} = \mathbf{0}.$

12. (a) Show that $\mathbf{F} = (2xy + z^3)\mathbf{i} + x^2\mathbf{j} + 3xz^2\mathbf{k}$ is a conservative force field. (b) Find the scalar potential. (c) Find the work done in moving an object in this field from $(1, -2, 1)$ to $(3, 1, 4).$

- (a) From Problem 11, a necessary and sufficient condition that a force will be conservative is that $\text{curl } \mathbf{F} = \nabla \times \mathbf{F} = \mathbf{0}.$

$$\text{Now } \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy + z^3 & x^2 & 3xz^2 \end{vmatrix} = \mathbf{0}.$$

Thus $\mathbf{F}$ is a conservative force field.

(b) *First Method.*

By Problem 10, $\mathbf{F} = \nabla\phi$ or $\frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k} = (2xy + z^3)\mathbf{i} + x^2\mathbf{j} + 3xz^2\mathbf{k}$. Then

$$(1) \frac{\partial\phi}{\partial x} = 2xy + z^3 \quad (2) \frac{\partial\phi}{\partial y} = x^2 \quad (3) \frac{\partial\phi}{\partial z} = 3xz^2$$

Integrating, we find from (1), (2) and (3) respectively,

$$\phi = x^2y + xz^3 + f(y, z)$$

$$\phi = x^2y + g(x, z)$$

$$\phi = xz^3 + h(x, y)$$

These agree if we choose $f(y, z) = 0$, $g(x, z) = xz^3$, $h(x, y) = x^2y$ so that $\phi = x^2y + xz^3$ to which may be added any constant.

Second Method.

Since $\mathbf{F}$ is conservative, $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of the path C joining (x_1, y_1, z_1) and (x, y, z) . Using the method of Problem 11(b),

$$\begin{aligned} \phi(x, y, z) &= \int_{x_1}^x (2xy_1 + z_1^3) dx + \int_{y_1}^y x^2 dy + \int_{z_1}^z 3xz^2 dz \\ &= (x^2y_1 + xz_1^3) \Big|_{x_1}^x + x^2y \Big|_{y_1}^y + xz^3 \Big|_{z_1}^z \\ &= x^2y_1 + xz_1^3 - x_1^2y_1 - x_1z_1^3 + x^2y - x^2y_1 + xz^3 - xz_1^3 \\ &= x^2y + xz^3 - x_1^2y_1 - x_1z_1^3 = x^2y + xz^3 + \text{constant} \end{aligned}$$

Third Method. $\mathbf{F} \cdot d\mathbf{r} = \nabla\phi \cdot d\mathbf{r} = \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz = d\phi$

Then
$$\begin{aligned} d\phi &= \mathbf{F} \cdot d\mathbf{r} = (2xy + z^3)dx + x^2dy + 3xz^2dz \\ &= (2xy dx + x^2 dy) + (z^3 dx + 3xz^2 dz) \\ &= d(x^2y) + d(xz^3) = d(x^2y + xz^3) \end{aligned}$$

and $\phi = x^2y + xz^3 + \text{constant}$.

(c) Work done $= \int_{P_1}^{P_2} \mathbf{F} \cdot d\mathbf{r}$

$$\begin{aligned} &= \int_{P_1}^{P_2} (2xy + z^3) dx + x^2 dy + 3xz^2 dz \\ &= \int_{P_1}^{P_2} d(x^2y + xz^3) = x^2y + xz^3 \Big|_{P_1}^{P_2} = x^2y + xz^3 \Big|_{(1, -2, 1)}^{(3, 1, 4)} = 202 \end{aligned}$$

Another Method.

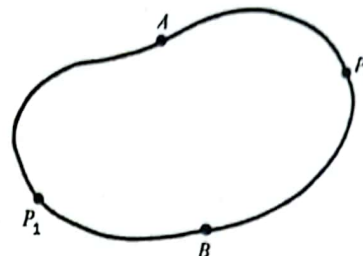
From part (b), $\phi(x, y, z) = x^2y + xz^3 + \text{constant}$.

Then work done $= \phi(3, 1, 4) - \phi(1, -2, 1) = 202$.

13. Prove that if $\int_{P_1}^{P_2} \mathbf{F} \cdot d\mathbf{r}$ is independent of the path joining any two points P_1 and P_2 in a given region, then $\oint \mathbf{F} \cdot d\mathbf{r} = 0$ for all closed paths in the region and conversely.

Let $P_1AP_2BP_1$ (see adjacent figure) be a closed curve. Then

$$\begin{aligned} \oint \mathbf{F} \cdot d\mathbf{r} &= \int_{P_1AP_2BP_1} \mathbf{F} \cdot d\mathbf{r} = \int_{P_1AP_2} \mathbf{F} \cdot d\mathbf{r} + \int_{P_2BP_1} \mathbf{F} \cdot d\mathbf{r} \\ &= \int_{P_1AP_2} \mathbf{F} \cdot d\mathbf{r} - \int_{P_1BP_2} \mathbf{F} \cdot d\mathbf{r} = 0 \end{aligned}$$



since the integral from P_1 to P_2 along a path through A is the same as that along a path through B , by hypothesis.

Conversely if $\oint \mathbf{F} \cdot d\mathbf{r} = 0$, then

$$\int_{P_1AP_2BP_1} \mathbf{F} \cdot d\mathbf{r} = \int_{P_1AP_2} \mathbf{F} \cdot d\mathbf{r} + \int_{P_2BP_1} \mathbf{F} \cdot d\mathbf{r} = \int_{P_1AP_2} \mathbf{F} \cdot d\mathbf{r} - \int_{P_1BP_2} \mathbf{F} \cdot d\mathbf{r} = 0$$

so that,
$$\int_{P_1AP_2} \mathbf{F} \cdot d\mathbf{r} = \int_{P_1BP_2} \mathbf{F} \cdot d\mathbf{r}.$$

14. (a) Show that a necessary and sufficient condition that $F_1 dx + F_2 dy + F_3 dz$ be an exact differential is that $\nabla \times \mathbf{F} = 0$ where $\mathbf{F} = F_1 \mathbf{i} + F_2 \mathbf{j} + F_3 \mathbf{k}$.
 (b) Show that $(y^2 z^3 \cos x - 4x^3 z) dx + 2z^3 y \sin x dy + (3y^2 z^2 \sin x - x^4) dz$ is an exact differential of a function ϕ and find ϕ .

- (a) Suppose $F_1 dx + F_2 dy + F_3 dz = d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz$, an exact differential. Then since x, y and z are independent variables,

$$F_1 = \frac{\partial \phi}{\partial x}, \quad F_2 = \frac{\partial \phi}{\partial y}, \quad F_3 = \frac{\partial \phi}{\partial z}$$

and so $\mathbf{F} = F_1 \mathbf{i} + F_2 \mathbf{j} + F_3 \mathbf{k} = \frac{\partial \phi}{\partial x} \mathbf{i} + \frac{\partial \phi}{\partial y} \mathbf{j} + \frac{\partial \phi}{\partial z} \mathbf{k} = \nabla \phi$. Thus $\nabla \times \mathbf{F} = \nabla \times \nabla \phi = 0$.

Conversely if $\nabla \times \mathbf{F} = 0$ then by Problem 11, $\mathbf{F} = \nabla \phi$ and so $\mathbf{F} \cdot d\mathbf{r} = \nabla \phi \cdot d\mathbf{r} = d\phi$, i.e. $F_1 dx + F_2 dy + F_3 dz = d\phi$, an exact differential.

- (b) $\mathbf{F} = (y^2 z^3 \cos x - 4x^3 z) \mathbf{i} + 2z^3 y \sin x \mathbf{j} + (3y^2 z^2 \sin x - x^4) \mathbf{k}$ and $\nabla \times \mathbf{F}$ is computed to be zero, so that by part (a)

$$(y^2 z^3 \cos x - 4x^3 z) dx + 2z^3 y \sin x dy + (3y^2 z^2 \sin x - x^4) dz = d\phi$$

By any of the methods of Problem 12 we find $\phi = y^2 z^3 \sin x - x^4 z + \text{constant}$.

15. Let $\mathbf{F}$ be a conservative force field such that $\mathbf{F} = -\nabla \phi$. Suppose a particle of constant mass m to move in this field. If A and B are any two points in space, prove that

$$\phi(A) + \frac{1}{2} m v_A^2 = \phi(B) + \frac{1}{2} m v_B^2$$

where v_A and v_B are the magnitudes of the velocities of the particle at A and B respectively.

$$\mathbf{F} = m\mathbf{a} = m \frac{d^2 \mathbf{r}}{dt^2}. \quad \text{Then } \mathbf{F} \cdot \frac{d\mathbf{r}}{dt} = m \frac{d\mathbf{r}}{dt} \cdot \frac{d^2 \mathbf{r}}{dt^2} = \frac{m}{2} \frac{d}{dt} \left(\frac{d\mathbf{r}}{dt} \right)^2.$$

$$\text{Integrating, } \int_A^B \mathbf{F} \cdot d\mathbf{r} = \frac{m}{2} v^2 \Big|_A^B = \frac{1}{2} m v_B^2 - \frac{1}{2} m v_A^2.$$

$$\text{If } \mathbf{F} = -\nabla\phi, \quad \int_A^B \mathbf{F} \cdot d\mathbf{r} = - \int_A^B \nabla\phi \cdot d\mathbf{r} = - \int_A^B d\phi = \phi(A) - \phi(B).$$

$$\text{Then } \phi(A) - \phi(B) = \frac{1}{2} m v_B^2 - \frac{1}{2} m v_A^2 \text{ and the result follows.}$$

$\phi(A)$ is called the *potential energy* at A and $\frac{1}{2} m v_A^2$ is the *kinetic energy* at A . The result states that the total energy at A equals the total energy at B (conservation of energy). Note the use of the minus sign in $\mathbf{F} = -\nabla\phi$.

16. If $\phi = 2xyz^2$, $\mathbf{F} = xy\mathbf{i} - z\mathbf{j} + x^2\mathbf{k}$ and C is the curve $x=t^2$, $y=2t$, $z=t^3$ from $t=0$ to $t=1$, evaluate the line integrals (a) $\int_C \phi d\mathbf{r}$, (b) $\int_C \mathbf{F} \times d\mathbf{r}$.

$$\begin{aligned} \text{(a) Along } C, \quad \phi &= 2xyz^2 = 2(t^2)(2t)(t^3)^2 = 4t^9, \\ \mathbf{r} &= x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = t^2\mathbf{i} + 2t\mathbf{j} + t^3\mathbf{k}, \text{ and} \\ d\mathbf{r} &= (2t\mathbf{i} + 2\mathbf{j} + 3t^2\mathbf{k}) dt. \text{ Then} \end{aligned}$$

$$\begin{aligned} \int_C \phi d\mathbf{r} &= \int_{t=0}^1 4t^9 (2t\mathbf{i} + 2\mathbf{j} + 3t^2\mathbf{k}) dt \\ &= \mathbf{i} \int_0^1 8t^{10} dt + \mathbf{j} \int_0^1 8t^9 dt + \mathbf{k} \int_0^1 12t^{11} dt = \frac{8}{11}\mathbf{i} + \frac{4}{5}\mathbf{j} + \mathbf{k} \end{aligned}$$

$$\text{(b) Along } C, \quad \mathbf{F} = xy\mathbf{i} - z\mathbf{j} + x^2\mathbf{k} = 2t^3\mathbf{i} - t^3\mathbf{j} + t^4\mathbf{k}.$$

$$\text{Then } \mathbf{F} \times d\mathbf{r} = (2t^3\mathbf{i} - t^3\mathbf{j} + t^4\mathbf{k}) \times (2t\mathbf{i} + 2\mathbf{j} + 3t^2\mathbf{k}) dt$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2t^3 & -t^3 & t^4 \\ 2t & 2 & 3t^2 \end{vmatrix} dt = [(-3t^5 - 2t^4)\mathbf{i} + (2t^5 - 6t^5)\mathbf{j} + (4t^3 + 2t^4)\mathbf{k}] dt$$

$$\begin{aligned} \text{and } \int_C \mathbf{F} \times d\mathbf{r} &= \mathbf{i} \int_0^1 (-3t^5 - 2t^4) dt + \mathbf{j} \int_0^1 (-4t^5) dt + \mathbf{k} \int_0^1 (4t^3 + 2t^4) dt \\ &= -\frac{9}{10}\mathbf{i} - \frac{2}{3}\mathbf{j} + \frac{7}{5}\mathbf{k} \end{aligned}$$

SURFACE INTEGRALS.

17. Give a definition of $\iint_S \mathbf{A} \cdot \mathbf{n} dS$ over a surface S in terms of limit of a sum.

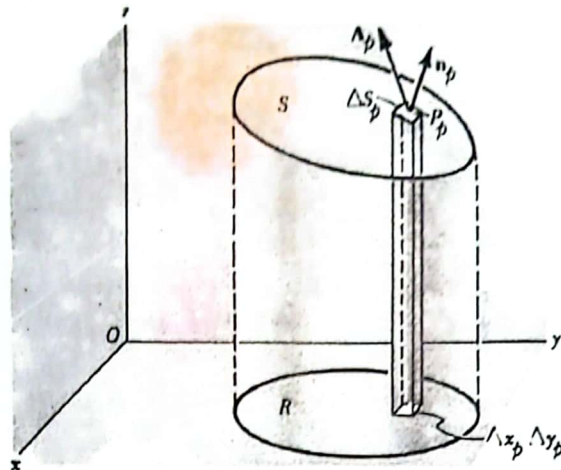
Subdivide the area S into M elements of area ΔS_p where $p = 1, 2, 3, \dots, M$. Choose any point P_p within ΔS_p whose coordinates are (x_p, y_p, z_p) . Define $\mathbf{A}(x_p, y_p, z_p) = \mathbf{A}_p$. Let $\mathbf{n}_p$ be the positive unit normal to ΔS_p at P . Form the sum

$$\sum_{p=1}^M \mathbf{A}_p \cdot \mathbf{n}_p \Delta S_p$$

where $\mathbf{A}_p \cdot \mathbf{n}_p$ is the normal component of $\mathbf{A}_p$ at P_p .

Now take the limit of this sum as $M \rightarrow \infty$ in such a way that the largest dimension of each ΔS_p approaches zero. This limit, if it exists, is called the surface integral of the normal component of $\mathbf{A}$ over S and is denoted by

$$\iint_S \mathbf{A} \cdot \mathbf{n} dS$$



18. Suppose that the surface S has projection R on the xy plane (see figure of Prob.17). Show that

$$\iint_S \mathbf{A} \cdot \mathbf{n} dS = \iint_R \mathbf{A} \cdot \mathbf{n} \frac{dx dy}{|\mathbf{n} \cdot \mathbf{k}|}$$

By Problem 17, the surface integral is the limit of the sum

$$(1) \quad \sum_{p=1}^M \mathbf{A}_p \cdot \mathbf{n}_p \Delta S_p$$

The projection of ΔS_p on the xy plane is $|(\mathbf{n}_p \Delta S_p) \cdot \mathbf{k}|$ or $|\mathbf{n}_p \cdot \mathbf{k}| \Delta S_p$ which is equal to $\Delta x_p \Delta y_p$ so that $\Delta S_p = \frac{\Delta x_p \Delta y_p}{|\mathbf{n}_p \cdot \mathbf{k}|}$. Thus the sum (1) becomes

$$(2) \quad \sum_{p=1}^M \mathbf{A}_p \cdot \mathbf{n}_p \frac{\Delta x_p \Delta y_p}{|\mathbf{n}_p \cdot \mathbf{k}|}$$

By the fundamental theorem of integral calculus the limit of this sum as $M \rightarrow \infty$ in such a manner that the largest Δx_p and Δy_p approach zero is

$$\iint_R \mathbf{A} \cdot \mathbf{n} \frac{dx dy}{|\mathbf{n} \cdot \mathbf{k}|}$$

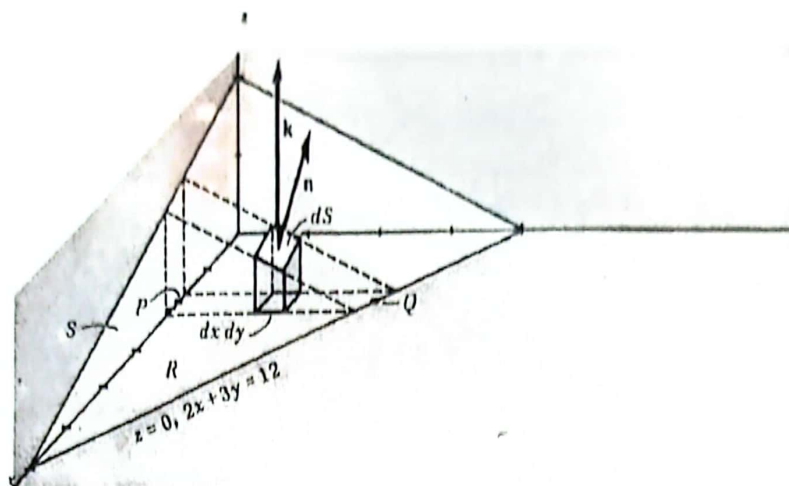
and so the required result follows.

Strictly speaking, the result $\Delta S_p = \frac{\Delta x_p \Delta y_p}{|\mathbf{n}_p \cdot \mathbf{k}|}$ is only approximately true but it can be shown on closer examination that they differ from each other by infinitesimals of order higher than $\Delta x_p \Delta y_p$, and using this the limits of (1) and (2) can in fact be shown equal.

19. Evaluate $\iint_S \mathbf{A} \cdot \mathbf{n} dS$, where $\mathbf{A} = 18x\mathbf{i} - 12y\mathbf{j} + 3y\mathbf{k}$ and S is that part of the plane

$2x + 3y + 6z = 12$ which is located in the first octant.

The surface S and its projection R on the xy plane are shown in the figure below.



From Problem 17,

$$\iint_S \mathbf{A} \cdot \mathbf{n} \, dS = \iint_R \mathbf{A} \cdot \mathbf{n} \frac{dx \, dy}{|\mathbf{n} \cdot \mathbf{k}|}$$

To obtain $\mathbf{n}$ note that a vector perpendicular to the surface $2x + 3y + 6z = 12$ is given by $\nabla(2x + 3y + 6z) = 2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}$ (see Problem 5 of Chapter 4). Then a unit normal to any point of S (see figure above) is

$$\mathbf{n} = \frac{2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}}{\sqrt{2^2 + 3^2 + 6^2}} = \frac{2}{7}\mathbf{i} + \frac{3}{7}\mathbf{j} + \frac{6}{7}\mathbf{k}$$

Thus $\mathbf{n} \cdot \mathbf{k} = (\frac{2}{7}\mathbf{i} + \frac{3}{7}\mathbf{j} + \frac{6}{7}\mathbf{k}) \cdot \mathbf{k} = \frac{6}{7}$ and so $\frac{dx \, dy}{|\mathbf{n} \cdot \mathbf{k}|} = \frac{7}{6} dx \, dy$.

$$\text{Also } \mathbf{A} \cdot \mathbf{n} = (18z\mathbf{i} - 12z\mathbf{j} + 3y\mathbf{k}) \cdot (\frac{2}{7}\mathbf{i} + \frac{3}{7}\mathbf{j} + \frac{6}{7}\mathbf{k}) = \frac{36z - 36 + 18y}{7} = \frac{36 - 12x}{7},$$

using the fact that $z = \frac{12 - 2x - 3y}{6}$ from the equation of S . Then

$$\iint_S \mathbf{A} \cdot \mathbf{n} \, dS = \iint_R \mathbf{A} \cdot \mathbf{n} \frac{dx \, dy}{|\mathbf{n} \cdot \mathbf{k}|} = \iint_R (\frac{36 - 12x}{7}) \frac{7}{6} dx \, dy = \iint_R (6 - 2x) dx \, dy$$

To evaluate this double integral over R , keep x fixed and integrate with respect to y from $y = 0$ (P in the figure above) to $y = \frac{12 - 2x}{3}$ (Q in the figure above); then integrate with respect to x from $x = 0$ to $x = 6$. In this manner R is completely covered. The integral becomes

$$\int_{x=0}^6 \int_{y=0}^{(12-2x)/3} (6 - 2x) \, dy \, dx = \int_{x=0}^6 (24 - 12x + \frac{4x^2}{3}) \, dx = 24$$

If we had chosen the positive unit normal $\mathbf{n}$ opposite to that in the figure above, we would have obtained the result -24 .

20. Evaluate $\iint_S \mathbf{A} \cdot \mathbf{n} \, dS$, where $\mathbf{A} = z\mathbf{i} + x\mathbf{j} - 3y^2z\mathbf{k}$ and S is the surface of the cylinder $x^2 + y^2 = 16$ included in the first octant between $z = 0$ and $z = 5$.

Project S on the xz plane as in the figure below and call the projection R . Note that the projection of S on the xy plane cannot be used here. Then

$$\iint_S \mathbf{A} \cdot \mathbf{n} \, dS = \iint_R \mathbf{A} \cdot \mathbf{n} \frac{dx \, dz}{|\mathbf{n} \cdot \mathbf{j}|}$$

A normal to $x^2 + y^2 = 16$ is $\nabla(x^2 + y^2) = 2xi + 2yj$. Thus the unit normal to S as shown in the adjoining figure, is

$$\mathbf{n} = \frac{2xi + 2yj}{\sqrt{(2x)^2 + (2y)^2}} = \frac{x\mathbf{i} + y\mathbf{j}}{4}$$

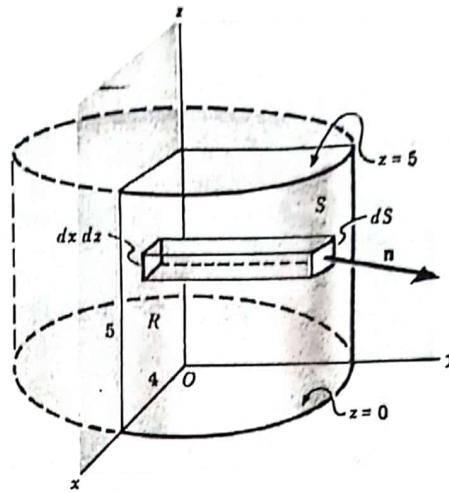
since $x^2 + y^2 = 16$ on S .

$$\mathbf{A} \cdot \mathbf{n} = (zi + xj - 3y^2zk) \cdot \left(\frac{x\mathbf{i} + y\mathbf{j}}{4}\right) = \frac{1}{4}(xz + xy)$$

$$\mathbf{n} \cdot \mathbf{j} = \frac{x\mathbf{i} + y\mathbf{j}}{4} \cdot \mathbf{j} = \frac{y}{4}$$

Then the surface integral equals

$$\iint_R \frac{xz + xy}{y} \, dx \, dz = \int_{z=0}^5 \int_{x=0}^4 \left(\frac{xz}{\sqrt{16-x^2}} + x \right) \, dx \, dz = \int_{z=0}^5 (4z + 8) \, dz = 90$$



21. Evaluate $\iint_S \phi \mathbf{n} \, dS$ where $\phi = \frac{3}{8}xyz$ and S is the surface of Problem 20.

We have
$$\iint_S \phi \mathbf{n} \, dS = \iint_R \phi \mathbf{n} \frac{dx \, dz}{|\mathbf{n} \cdot \mathbf{j}|}$$

Using $\mathbf{n} = \frac{x\mathbf{i} + y\mathbf{j}}{4}$, $\mathbf{n} \cdot \mathbf{j} = \frac{y}{4}$ as in Problem 20, this last integral becomes

$$\begin{aligned} \iint_R \frac{3}{8}xz(x\mathbf{i} + y\mathbf{j}) \, dx \, dz &= \frac{3}{8} \int_{z=0}^5 \int_{x=0}^4 (x^2z\mathbf{i} + xz\sqrt{16-x^2}\mathbf{j}) \, dx \, dz \\ &= \frac{3}{8} \int_{z=0}^5 \left(\frac{64}{3}z\mathbf{i} + \frac{64}{3}z\mathbf{j} \right) \, dz = 100\mathbf{i} + 100\mathbf{j} \end{aligned}$$

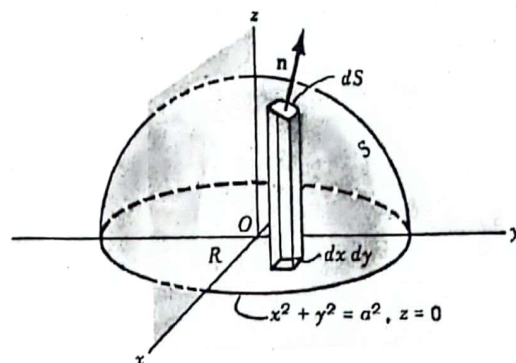
22. If $\mathbf{F} = y\mathbf{i} + (x - 2xz)\mathbf{j} - xy\mathbf{k}$, evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS$ where S is the surface of the sphere

$x^2 + y^2 + z^2 = a^2$ above the xy plane.

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & x - 2xz & -xy \end{vmatrix} = x\mathbf{i} + y\mathbf{j} - 2z\mathbf{k}$$

A normal to $x^2 + y^2 + z^2 = a^2$ is

$$\nabla(x^2 + y^2 + z^2) = 2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}$$



Then the unit normal $\mathbf{n}$ of the figure above is given by

$$\mathbf{n} = \frac{2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}}{\sqrt{4x^2 + 4y^2 + 4z^2}} = \frac{x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{a}$$

since $x^2 + y^2 + z^2 = a^2$.

The projection of S on the xy plane is the region R bounded by the circle $x^2 + y^2 = a^2$, $z = 0$ (see figure above). Then

$$\begin{aligned} \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS &= \iint_R (\nabla \times \mathbf{F}) \cdot \mathbf{n} \frac{dx \, dy}{|\mathbf{n} \cdot \mathbf{k}|} \\ &= \iint_R (x\mathbf{i} + y\mathbf{j} - 2z\mathbf{k}) \cdot \left(\frac{x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{a} \right) \frac{dx \, dy}{z/a} \\ &= \int_{x=-a}^a \int_{y=-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} \frac{3(x^2+y^2) - 2a^2}{\sqrt{a^2-x^2-y^2}} \, dy \, dx \end{aligned}$$

using the fact that $z = \sqrt{a^2 - x^2 - y^2}$. To evaluate the double integral, transform to polar coordinates (ρ, ϕ) where $x = \rho \cos \phi$, $y = \rho \sin \phi$ and $dy \, dx$ is replaced by $\rho \, d\rho \, d\phi$. The double integral becomes

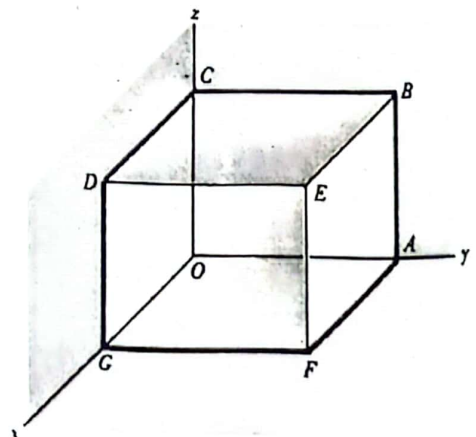
$$\begin{aligned} \int_{\phi=0}^{2\pi} \int_{\rho=0}^a \frac{3\rho^2 - 2a^2}{\sqrt{a^2 - \rho^2}} \rho \, d\rho \, d\phi &= \int_{\phi=0}^{2\pi} \int_{\rho=0}^a \frac{3(\rho^2 - a^2) + a^2}{\sqrt{a^2 - \rho^2}} \rho \, d\rho \, d\phi \\ &= \int_{\phi=0}^{2\pi} \int_{\rho=0}^a \left(-3\rho\sqrt{a^2 - \rho^2} + \frac{a^2\rho}{\sqrt{a^2 - \rho^2}} \right) d\rho \, d\phi \\ &= \int_{\phi=0}^{2\pi} \left[(a^2 - \rho^2)^{3/2} - a^2\sqrt{a^2 - \rho^2} \right]_{\rho=0}^a d\phi \\ &= \int_{\phi=0}^{2\pi} (a^3 - a^3) d\phi = 0 \end{aligned}$$

★ 23. If $\mathbf{F} = 4xz\mathbf{i} - y^2\mathbf{j} + yz\mathbf{k}$, evaluate $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$

where S is the surface of the cube bounded by $x=0$, $x=1$, $y=0$, $y=1$, $z=0$, $z=1$.

Face $DEFG$: $\mathbf{n}=\mathbf{i}$, $x=1$. Then

$$\begin{aligned} \iint_{DEFG} \mathbf{F} \cdot \mathbf{n} \, dS &= \int_0^1 \int_0^1 (4z\mathbf{i} - y^2\mathbf{j} + yz\mathbf{k}) \cdot \mathbf{i} \, dy \, dz \\ &= \int_0^1 \int_0^1 4z \, dy \, dz = 2 \end{aligned}$$



Face ABCO: $\mathbf{n} = -\mathbf{i}$, $x = 0$. Then

$$\iint_{ABCO} \mathbf{F} \cdot \mathbf{n} \, dS = \int_0^1 \int_0^1 (-y^2 \mathbf{j} + yz \mathbf{k}) \cdot (-\mathbf{i}) \, dy \, dz = 0$$

Face ABEF: $\mathbf{n} = \mathbf{j}$, $y = 1$. Then

$$\iint_{ABEF} \mathbf{F} \cdot \mathbf{n} \, dS = \int_0^1 \int_0^1 (4xz \mathbf{i} - \mathbf{j} + z \mathbf{k}) \cdot \mathbf{j} \, dx \, dz = \int_0^1 \int_0^1 -dx \, dz = -1$$

Face OGDC: $\mathbf{n} = -\mathbf{j}$, $y = 0$. Then

$$\iint_{OGDC} \mathbf{F} \cdot \mathbf{n} \, dS = \int_0^1 \int_0^1 (4xz \mathbf{i}) \cdot (-\mathbf{j}) \, dx \, dz = 0$$

Face BCDE: $\mathbf{n} = \mathbf{k}$, $z = 1$. Then

$$\iint_{BCDE} \mathbf{F} \cdot \mathbf{n} \, dS = \int_0^1 \int_0^1 (4x \mathbf{i} - y^2 \mathbf{j} + y \mathbf{k}) \cdot \mathbf{k} \, dx \, dy = \int_0^1 \int_0^1 y \, dx \, dy = \frac{1}{2}$$

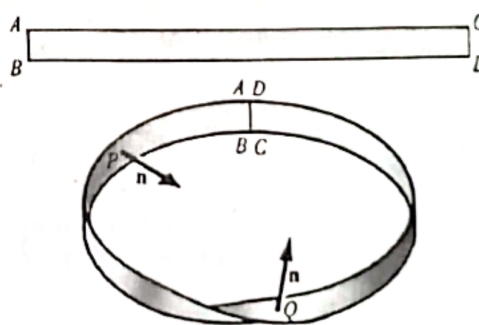
Face AFGO: $\mathbf{n} = -\mathbf{k}$, $z = 0$. Then

$$\iint_{AFGO} \mathbf{F} \cdot \mathbf{n} \, dS = \int_0^1 \int_0^1 (-y^2 \mathbf{j}) \cdot (-\mathbf{k}) \, dx \, dy = 0$$

$$\text{Adding, } \iint_S \mathbf{F} \cdot \mathbf{n} \, dS = 2 + 0 + (-1) + 0 + \frac{1}{2} + 0 = \frac{3}{2}.$$

24. In dealing with surface integrals we have restricted ourselves to surfaces which are two-sided. Give an example of a surface which is not two-sided.

Take a strip of paper such as $ABCD$ as shown in the adjoining figure. Twist the strip so that points A and B fall on D and C respectively, as in the adjoining figure. If $\mathbf{n}$ is the positive normal at point P of the surface, we find that as $\mathbf{n}$ moves around the surface it reverses its original direction when it reaches P again. If we tried to color only one side of the surface we would find the whole thing colored. This surface, called a *Möbius strip*, is an example of a one-sided surface. This is sometimes called a *non-orientable* surface. A two-sided surface is *orientable*.



VOLUME INTEGRALS

25. Let $\phi = 45x^2y$ and let V denote the closed region bounded by the planes $4x + 2y + z = 8$, $x = 0$, $y = 0$, $z = 0$. (a) Express $\iiint_V \phi \, dV$ as the limit of a sum. (b) Evaluate the integral in (a).